

Bonian Han

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Education

Boise State University

Fall 2026

Ph.D. in Computing (expected start: Fall 2026)

Advisor: Prof. Yu Zhang

New Jersey Institute of Technology

2025 - 2026

Ph.D. studies in Computer Science

Advisor: Prof. Zhi Wei | GPA: 4.0

Hangzhou Dianzi University

Sept 2021 - July 2025

B.S. in Statistics

GPA: 3.48 (84.8/95)

Research Interests

Computer vision; continual learning; embodied AI; trustworthy machine learning, including model calibration and uncertainty quantification; neural representation methods for spatial transcriptomics.

Publications

1. J. Zhang, **B. Han**, G. Liang, and Y. Zhang. "Rethinking Learned Occupancy in Autonomous Active Mapping with Observation-Gated Filtering." *IEEE/RSJ IROS 2026 Workshop on Space Exploration and Sustained Operations Beyond Earth*, 2026.
Oral; Best Paper Runner-Up Award.
2. **B. Han**, C. Qi, P. Musialski, and Z. Wei. "INST-Align: Implicit Neural Alignment for Spatial Transcriptomics via Canonical Expression Fields." *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. Accepted. Paper.
3. **B. Han**, Y. P. Masupalli, X. Xing, and G. Liang. "Improving Medical Imaging Model Calibration through Probabilistic Embedding." *IEEE International Conference on Big Data*, 2024.
doi:10.1109/BigData62323.2024.10825661.
4. **B. Han**, C. Moran, J. Yang, Y. Lee, Z. Cao, and G. Liang. "Multi-Scale Self-Supervised Consistency Training for Trustworthy Medical Imaging Classification." *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024. Podium (acceptance rate approx. 28%); H5-Index: 48. doi:10.1109/EMBC53108.2024.10782322.
5. J. Zulu, **B. Han**, I. Alsmadi, and G. Liang. "Contextualized Embedding-Based Approach for Enhanced SQL Injection Detection." *62nd ACM SouthEast Annual Conference*, 2024. doi:10.1145/3603287.3651187.

Research Experience

New Jersey Institute of Technology

Sept 2025 - 2026

Research with Prof. Zhi Wei

Conducted research on spatial transcriptomics and neural representation methods. Worked on INST-Align, coupling spatial alignment with a canonical expression field; accepted to MICCAI 2026.

Texas A&M University-San Antonio

Nov 2023 - Nov 2024

Research with Dr. Gongbo Liang

Conducted computer vision research on medical image analysis, model calibration, and feature extraction to reduce model bias. Work resulted in publications at IEEE EMBC, ACMSE, and IEEE Big Data.

Skills & Languages

Technical skills: Python, C/C++, PyTorch, SQL, R, Linux, Bash, Git, Docker

Languages / tests: TOEFL: 91; GRE: 326 (Verbal: 156, 69th percentile; Quantitative: 170, 92nd percentile)

Additional Experience

Ballkids Coach & Manager

Sept 2024

Hangzhou Open ATP 250 and ATP Challenger

Trained and managed professional ballkids (ages 9-14), coordinated on-court schedules across departments, and ensured smooth match operations, court etiquette, and safety protocols.

Tennis Ballkids & Ceremony Assistant

Sept - Oct 2023

19th Asian Games and Asian Para Games

Managed on-court operations and assisted players and umpires. Featured by China Central Television (CCTV); honored as an Outstanding and Excellent Volunteer.

Awards & Honors

Best Paper Runner-Up Award

2026

IEEE/RSJ IROS Workshop on Space Exploration and Sustained Operations Beyond Earth; "Rethinking Learned Occupancy in Autonomous Active Mapping with Observation-Gated Filtering." Oral presentation.

Outstanding & Excellent Volunteer, National, 2023

Second Prize, Statistics Research Competition, Provincial (State), 2023

Third Prize, National College Students Mathematics Competition (Online Challenge), 2022

Fifth Place, University Basketball Annual Competition, May 2022